

1. IDENTIFICATION

Finding and collecting potentially relevant studies

1.1 Formulate research questions

Define the scope, objectives, and research questions.

1.2 Select information sources

IEEE Xplore, Scopus, ACM Digital Library, SpringerLink, ScienceDirect, and Google Scholar.

1.3 Develop search strategy

Define keywords, synonyms, Boolean operators, and search strings.

1.4 Search literature

Run searches in the selected databases and identify additional studies through snowballing.

1.5 Remove duplicates

Consolidate retrieved records and remove duplicate publications.

Key outputs

- Research questions
- Search strategy
- Deduplicated record set

2. SCREENING

Filtering studies based on titles and abstracts

2.1 Title screening

Exclude records that are clearly outside the review scope.

2.2 Abstract screening

Apply the predefined inclusion and exclusion criteria.

2.3 Record exclusion decisions

Document the principal reasons for excluding records.

Key outputs

- Screened records
- Excluded records
- Potentially eligible studies

3. ELIGIBILITY

Assessing full-text studies for eligibility and quality

3.1 Retrieve full texts

Obtain the complete publications for potentially eligible studies.

3.2 Assess full-text eligibility

Apply the predefined inclusion and exclusion criteria.

3.3 Quality assessment

Evaluate methodological quality using the predefined quality checklist.

3.4 Record exclusion decisions

Document reasons for exclusion following full-text assessment.

Key outputs

- Full texts assessed
- Quality-assessed studies
- Eligible research corpus

4. INCLUDED STUDIES AND SYNTHESIS

Extracting data and synthesizing evidence

4.1 Data extraction and coding

Extract predefined variables and classify the included studies using a structured data-extraction framework.

4.2 Data synthesis

Perform descriptive, comparative, quantitative, and thematic synthesis.

4.3 Report findings

Present research characteristics, technical synthesis, research gaps, and future directions.

Key outputs

- Final research corpus
- Structured extraction dataset
- Research taxonomy
- Technical synthesis and research roadmap